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Developing a Prototype to Translate Text and Speech to Pakistan Sign Language With Bilingual Subtitles: A Framework

Journal of Educational Technology Systems
2018, Vol. 47(2) 248–266
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DOI: 10.1177/0047239518794168
journals.sagepub.com/home/ets
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Abstract

The purpose of the study is to provide a literature review of the work done on sign language (SL) around the world and in Pakistan and to develop a translation tool of speech and text to Pakistan Sign Language (PSL) with bilingual subtitles. Information and communication technology and tools development for teaching and learning purposes improve the learning process and facilitate both teachers and students. In Pakistan, unimpaired people face a lot of problems to communicate with deaf people due to the lack of SL understanding, learning resources, and interpreters. This problem is faced by the teachers who communicate with deaf students in the classrooms. The communication gap is filled with the development of a translation tool as Haseeb and Illyas concluded in their study that using this kind of tool, deaf people will have more opportunities to communicate with other members of society at every level. Different components of technology such as Python programming language, Natural Language Tool Kit, prerecorded PSL videos, Linux-based server, and databases are used to develop the prototype of PSL translation tool. This study provides a literature review to highlight the existing technological work done around the

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world and in Pakistan and also provides an architectural framework of the PSL translation tool which is developed by the researchers to facilitate the people who face difficulty to communicate with deaf people.

Keywords
sign language, Pakistan Sign Language, deaf, hard of hearing, information and communication technology

Introduction

Language is a mode of communication through which people interact with each other. They know, understand, and share their thoughts with each other with the help of language. According to the Oxford Dictionary, *language* is defined as follows: "The method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way" ("Language | Definition of language in English by Oxford Dictionaries," n.d.)." There are different languages spoken by the people to communicate with each other. There are some people who cannot speak, but they interact with each other using sign language (SL). Mostly, signs represent ideas and no single words. Some signs are iconic and use the visual image of something to represent that specific thing. SL varies from country to country and region to region just like spoken language. Well-developed countries have developed their standard SLs. They have all the necessary elements of SL, such as grammar, structure, and standard signs. Some of those countries are American Sign Language (ASL), British Sign Language, Arabic Sign Language, and Chinese Sign Language, and so forth.

Pakistan Sign Language (PSL) is under the development phase. Researchers in the field of SL are working to develop a more centralized and standardized SL in Pakistan. PSL is not fully developed yet. Different organizations are working on developing PSL, but still, according to researchers' best knowledge, they are unable to compile a standardized version. The following are the four major institutes that have a great role in developing PSL: Sir Syed Deaf Association (SDA), Rawalpindi; Anjuman Behbood-e-Samat-e-Atfal (ABSA), Karachi; National Institute of Special Education (NISE), Islamabad; and Pakistan Association of Deaf (PAD), Karachi.

Deaf people make up to 72 million of the world's population (Bukhari, Rehman, Malik, Kamboh, & Salman, 2015). They need a language to communicate with people around them. In Pakistan, according to the census of Pakistan 1998, there are 3,286,630 disabled people, and among those, 7.43% are deaf people. It is a major portion of overall population of Pakistan. These deaf or hard of hearing people often face a lot of difficulties while

communicating with other people in the society because there is no proper understanding of SL among people. Hence, the deaf community also faces some problems, for example, they cannot find a job to meet their financial needs. To fill this communication gap, technology can be used as a tool similar to the other developed countries. Developed countries have already developed some kind of tools/software that help deaf people to communicate with unimpaired people.

Different types of tools are available worldwide to translate speech or text to SL or vice versa. For example, there is a mobile application named "ProDeaf" which animate the text to ASL. Another tool S2V (Sign to Voice), developed by Foong, Low, and Wibowo (2008), can translate the signs into voice. These are few examples of tools that help deaf community and unimpaired people to communicate with each other. In Pakistan, researchers have also worked in this field, for example, "SignSpeak" by Bukhari et al. (2015); recognition of gestures through fuzzy classifiers by Kausar, Javed, and Sohail (2008); speech translation into PSL by Haseeb and Ilyas (2012); and a proposed framework by Khan et al. (2015) to translate English and Urdu speech and text to PSL animation. There is a need to bridge the communication gap between hard of hearing and unimpaired people in the Pakistani community. So, this study provides the framework of a translation tool to translate English speech and text to PSL with bilingual (English and Urdu) subtitles to bridge the communication gap between deaf and unimpaired people. Bilingual (English and Urdu) subtitles are used to enhance the vocabulary of deaf people.

Literature Review

The following literature review details the history and development of PSL and technological work that has done around the world and in Pakistan to bridge the communication gap between deaf and unimpaired people. It also shows the need for the development of speech and text to PSL translation tool and the viability of bilingual (English and Urdu) subtitles to enhance the vocabulary of deaf students.

Different organizations are working on developing PSL, but they are still unable to compile a standardized version. PSL is not fully developed yet. Four major institutes play a major role in developing PSL: SDA, Rawalpindi; ABSA, Karachi; NISE, Islamabad; and PAD, Karachi. SDA published first PSL dictionary. As Sulman & Zuberi (2000) said in the synopsis of PSL, the dictionary contained 750 signs that were used only in Rawalpindi region.

ABSA is another major contributor in the development of PSL. This group standardizes and documents the PSL. The research group of ABSA published the following different notable booklets:

a. A Dictionary of Pakistan Sign Language
b. Relationships in Sign Language
c. Time and Seasons in Sign Language

d. The Anatomy and Body Actions in Sign Language
e. Numeration in Sign Language
f. A series of stories
g. Interacting development of new signs with teachers

NISE established by state in 1976 has the responsibility of manpower development for centers of education throughout Pakistan. The main focus of NISE was to use already-developed signs that are region-based to develop a common SL that is accepted by all deaf communities in Pakistan.

PAD is a privately administered nongovernmental organization in Karachi, Pakistan. The main focus of PAD was also to develop a SL, which is accepted by all deaf communities across Pakistan. Their research group has published four books after discussion and negotiation. Those books are as follows:

a. Workbook of Alphabet signs in Urdu
b. Workbook of Alphabet signs in English
c. 500-word dictionary with new words and modified signs
d. Traffic signs for Deaf drivers

Family Educational Services Foundation (FESF) is a nonprofit educational organization working in Pakistan. It has a dedicated program for deaf with the name of "Deaf Reach" program. FESF has created and documented a digital lexicon of 5,000 words in English, Urdu, and PSL. They have also developed a mobile app which also contains 5,000 words. They have published a book as well: *Pakistan Sign Language 1000 Basic Signs*. FESF also has training programs that are designed to improve the communication abilities of teachers and families of the deaf based on lexicons of PSL.

Across the globe, a lot of development is taking place on SL. Researchers are working to fill the communication gap between hard of hearing and unimpaired people. Technology is the key contributor in the process of reducing this communication gap. Some well-developed countries that are technologically advanced have already developed some tools to translate speech and text to SL and vice versa. Some countries are in the process of developing this kind of tools to meet the needs of communication between impaired and unimpaired people.

A research was conducted by Tejedor, López, Bolaños, and Colás (2006) in Spain to allow the hard of hearing people to communicate properly with other people. The aim of that research was to help the hard of hearing and unimpaired people to communicate with each other without learning the SL. They developed a tool that uses 3G mobile device and video calling feature. This tool uses the unimpaired person's voice and after that shows it in the video call after translating it to SL.

SL to voice translation system can be a state-of-the-art tool that can translate the SL to the voice. Most of the tools enable unimpaired people to communicate

with deaf. But, such kind of tool can translate sign to the voice. A research was conducted by Foong et al. (2008) to develop a prototype that can translate the signs to voice (S2V). This system was developed using "Feed Forward Neural Network" for sign detection system. To develop this prototype, sets of universal hand signs recorded with video camera, these videos were used to train this artificial intelligent tool. The neural network system showed satisfactory results after developing and testing it.

The world is working in the way to pave the way for communication between disabled people, for example, deaf and blind. Across the globe, a lot of research is going on with the help of state or private organizations or individual basis. As technology, nowadays, advances, experts in technology can mold it for the betterment of humanity. Moustakas et al. (2006) developed "multimodal tools and interfaces for the inter-communication between visually impaired and deaf and mute people" (Moustakas et al., 2006). The researchers developed this tool during the "eNTERFACE-2006" summer workshop to enable the intercommunication between hard of hearing and blind people. It is a state-of-the-art tool. A set of technologies were integrated into a treasure hunting game application. The use of this game into the multimodal communication tool is for the sake of entertainment as well as learning and communication. There are different technologies used by the researchers to get the desired outcome. Those technologies are gesture recognition, haptics, speech analysis and synthesis, and SL analysis and synthesis. This tool was tested during the same workshop in which it was developed, "eNTEFACE." It was tested among the two users, although it was a limited tool, and it provided the satisfactory results. A similar study was conducted by Noberto et al. (2015) in Portugal. These Portuguese researchers developed a game to teach the SL. They used the game in their tool because it increases the attention of the learners and people learn faster than other traditional ways. The main focus of this research is to develop a bidirectional Portuguese sign language translator. This tool helps to optimize the learning and communication using the Portuguese sign language: "The project is directed to all those that want to improve or learn their sign language skills" (Noberto et al., 2015, p. 267).

World is moving toward more advancement in technology. It is influencing every field of life, and experts are molding technology for the benefit of human kind. SL translation tools are also advancing, and researchers are developing 3D avatars to show the translated signs from the text. They are also working on the accuracy of translated version of the SL. A study was conducted by Othman and Jemni (2011) in Tunisia to translate English written text to ASL. In this research, they used a modified Mosses tool and "results are synthesized through a 3D avatar for interpretation" (Othman & Jemni, 2011, p. 65). This tool was developed after several experiments with English to ASL statistical SL machine translation (MT). The aim of this tool is to help interpreters, unimpaired people, and deaf or hard of hearing education system. A similar research conducted by

the National Technical Institute for the Deaf in 2012 puts the light on 3D avatar system used in SL translation tool, in which, they are working on the 3D avatar system and also focusing on the importance of avatars. Researcher says that a lot of notable progress can be seen in the development of avatar system. They concluded that some more advancement in linguistics and MT is required to develop an ideal and error-free translation system that can be fully embraced by the hard of hearing community.

In India, some notable work is done by the researchers to bridge the communication gap between hard of hearing and unimpaired people. They have used technology to make the life of impaired people easy. The biggest barrier that the deaf community faces is the communication gap with people around them. Sankar Kumar, Jenitha, Narmadha, and Suganya (2014) have developed a virtual tongue for communication purposes. The project is done by the researchers to make deaf people self-reliable and independent while communicating with other people. This system is designed to produce speech sound for the people who are deaf. "Based on the request from the user by pressing the icons thereby this module deserves inarticulate people" (Sankar Kumar et al., 2014, p. 155). This system was developed with USB port, speakers for audio input, and a remote control with icons on it.

As world is advancing in technology, Pakistan is also keeping pace with the advancement of technology and modern world. As world has created a lot of tools to bridge the communication gap between unimpaired and impaired people, Pakistani researchers are also working on it. A study was conducted by Mehdi and Khan (2002) to develop a sensory glove for the translation of ASL into text form. They introduced this concept to make the communication easy between deaf and normal people. This remarkable project is known as "Talking Hands." Artificial neural network was used by the researchers to recognize the values coming from sensory glove. "These values are then categorized in 24 alphabets of English language and two punctuation symbols introduced by the author. So, mute people can write complete sentences using this application" (Mehdi & Khan, 2002, p. 2204). Another similar study was conducted by Bukhari et al. (2015) to make the communication between deaf and unimpaired people easy. Researchers developed a sensory glove to translate the ASL into speech. The glove had flex and contact sensors to detect the movement of finger to convert it to speech. It was a very well-developed tool with 92% accuracy rate. It was initially able to convert the alphabets into text and then speech. Another similar research was conducted by Raziq and Latif (2016) to translate the PSL into text form. They proposed a sign recognition system for PSL using the leap motion device technology. It consists of two modules for training a communication module. First module trains the system to translate PSL, and the other module, which is "communication module," collects the data using leap motion device and algorithms working on the backend to detect and recognize the sign to translate into text form.

Fatima and Huma (2011) conducted a study in Pakistan which is based on Urdu alphabets in PSL. This tool analyzes the signs and displays the text related to those signs on a computer interface. In this system, static signs were considered by the researchers. According to the researchers, it is the first work done for the recognition of PSL without the use of any type of glove . It is a vision-based system for the people who want to use or uses PSL in their life.

Some other advancement in technological work to bridge the communication gap between deaf and unimpaired people is taking place in Pakistan. Kausar et al. (2008) created a fuzzy classifier to recognize the signs made by deaf people. Colored gloves are used in this method to identify each fingertip and joint to translate. The accuracy rate of the system was 95%.

Researchers across the Pakistan are paying attention to develop more advanced PSL translation tools which can translate the sign from text and speech. Khan et al. (2015) highlighted the challenges in the process of developing a tool/system for SL translation purposes in Pakistan. In a study, they proposed an architectural framework that can help in translating English or Urdu text/speech into animated form of PSL and vice versa (Khan et al., 2015).

A prototype was developed by Haseeb and Illyas (2012) to translate the speech into PSL. The study was conducted to explore the speech recognition and MT techniques, to design and develop a generic and automated system, and to convert speech to PSL.

In education sector, information and communication technology (ICT) is already integrated in the developed countries, and it is paving its way to be integrated in developing countries' educational settings. All fragments of life are almost connected with technology, for example, education, medical, agriculture, defense, art and culture, and so on. Technology is making its way into the lives of people and providing them an easy and comfortable life. Use of technology in education is very beneficial for students and teachers. A study conducted by Mullamaa (2010) showed the benefits of using technology in the learning process. This study focuses on the technological environment in classrooms which helps students to motivate themselves for learning process and helps to create a student-centered learning environment. The researcher also analyzes the solutions provided by ICT for the classroom tasks such as group work, pair work, and developing a personal opinion about any specific topic. Findings showed that ICT integration assists teachers in classrooms, "build trust" between teacher and student, and "creating the feeling of belonging together." It develops a classroom that is less teacher-centered and more student-centered (Mullamaa, 2010). Another similar research conducted by Padurean and Margan (2009) concluded that conventional classrooms and learning activities can be improved using technology in teaching and learning process. Ullah et al. (2016) conducted a research in Pakistan to check the outcome of integration of technology in the classrooms. Computer-based tools are cutting edge technology which is changing the face of conventional education

system and process. Researchers implemented an interactive white board in classroom for teaching purposes. This implementation of technology in classroom showed that it has increased the attraction level of students toward learning process.

The translation tools that are discussed earlier can be used in educational setting of Pakistan to bridge the communication gap between deaf students and unimpaired teachers who have lesser knowledge of SL. It can also make the deaf students motivate toward the learning process and enhance their knowledge.

Architectural Framework

This research is presenting an architectural framework of PSL translation tool developed by the researchers to translate the English text and speech to PSL. It can convert the speech and text to PSL with the help of prerecorded videos with bilingual subtitles. An architecture of the framework has been presented in Figure 1. In this figure, researchers presented the main steps that take place when the input (speech or text) is provided by the user in the PSL translation tool. Pre-recorded videos are used by the researchers to show the output of text or speech in video form. There are two types of inputs accepted by the translation tool, that is, English text and English speech. These inputs further go through the different phases and compile one video to translate English sentences into PSL. The detailed algorithm of translation mechanism is provided in the Figure 2.

Tool Specifications

There are different technological tools and frameworks used to develop the PSL translation tool.

Programming Language

Python is known as object-oriented and high-level programming language with dynamic structure. According to the official website of python, "Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together" ("What is Python?," n.d.). Python is a simple and easy to learn language. It has a large number of libraries available in the source binary. For the development of PSL translation tool, Python programming language is used and the version 3.6.4 is used.

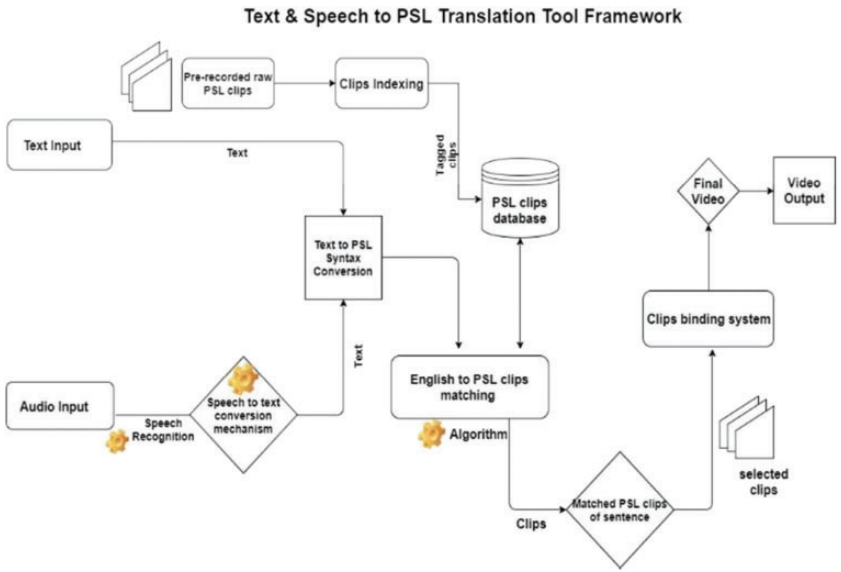


Figure 1. Text and speech to Pakistan sign language translation tool framework. In this figure, researchers presented the main steps which will take place in the development of PSL translation tool. Prerecorded videos will be used by the researchers to show the output of text or speech in video form. There are two types of inputs accepted by the translation tool, that is, English text and English speech. These inputs will further go through the different phases and compile one video to translate English sentences into PSL.

Django (Web Framework)

As the current tool is web based, therefore, there was a need to use a framework of Python that supports developing the application on web platform. Django is the best framework to develop web-based application that uses the database-driven website or application. As PSL translation tool is using a large corpus of PSL video, so the researcher used Django web framework.

PyMongo

Database is an important part of the whole application development. This study has a large number of PSL videos, stored in its database. Therefore, there is a need to select and use the database carefully. In this study, "MongoDB" database is used to store and process data. PyMongo is used in this study. It is basically a distribution of Python that has tools to work with MongoDB. It is recommended to work with MongoDB from Python.

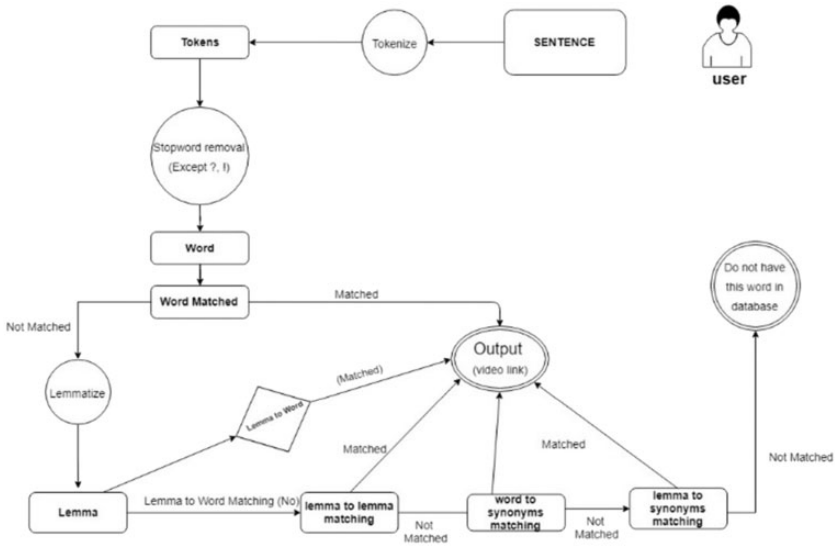


Figure 2. Clip matching algorithm. A detailed diagram of the clip matching algorithm is given below which is explaining all the necessary steps which will take place in the clip matching mechanism and produce a translation in Pakistan Sign Language.

PyDictionary

There were limited number of PSL videos in the database. Each word has one video, but there are some other words that can also be used in the place of a word. To expand the database, researcher used "PyDictionary" for Python to get meanings, synonyms, antonyms, and translations. In the case of PSL translation tool, research will only use the synonyms option to increase number of words in the corpus. PyDictionary module uses "WordNet" to get meaning, "thesaurus.com" to get synonyms, and "Google" for translation purposes.

MoviePy

A large number of video clips exist in the database of PSL translation tool. One word has one video clip which is separate from other video clips. When the user writes a sentence in the text area of the translation tool, it fetches the video for each word from the database. Now, there is a need to bind all the videos and form one video clip to show the output of a written sentence. Researcher used "MoviePy" for this purpose. MoviePy is a Python module for video editing, which can be used for basic operations (such as cuts, concatenations, and title insertions), video compositing (a.k.a. nonlinear editing), video processing, or to create

advanced effects. It can read and write the most common video formats, including GIF.

Natural Language Tool Kit

Natural Language Tool Kit (NLTK) is a suite of libraries to work on natural language processing for English developed in Python programming language. It is called an amazing tool for teaching and working in computational linguistics. Under the umbrella of NLTK, there are different libraries that are used to process the input.

Tokenized Words and Sentences

Words and sentences tokenization process takes place when word or sentence tokenize library is used by the developer while working on Python under NLTK. It is used to tokenize the sentences in separate words to better match the videos in the database. A sentence or data can be split into words using the method "sent_tokenize()." Library under discussion is used by the researcher to tokenize the sentences.

Identifying Bigrams and Trigrams

Due to the previous step which is tokenization, all the words in a sentence tokenizes, but there are few words that provide meaning when connected with another word in a sentence. For example, "white" and "snow" becomes different word when tokenizing a sentence, but these two words should come together. Here comes the identification of bigrams or trigrams; this library of NLTK connects the words that are connected with each other. The researcher used this library to identify bigrams and trigrams in the sentences that need to be translated with the help of PSL translation tool.

Stop Word Removal

The stop word removal library in NLTK helps to remove the words that do not confer much semantic value, for example, the, it, a, and so on. These words, while translating in PSL, do not convey much meanings and also do not have the videos in the database. So, to remove such kind of words, NLTK library is used by the researchers.

Lemmatizing

Words come in different morphological forms, such as derivational morphemes and inflectional morphemes. While developing a PSL translation tool, there is a

need to normalize the corpus. So the lemmatization library is being used in this tool development process. The goal of lemmatization is to reduce inflectional forms and sometimes derivationally related forms of a word to a common base form. It usually refers for doing things properly with the use of a vocabulary and morphological analysis of words, normally aiming to remove inflectional endings only and to return the base or dictionary form of a word, which is known as the lemma.

Tool Development Process

The development process of speech and text to PSL translation tool is described as follows:

PSL Clips Database Development

To provide the output of a written sentence in PSL, there is a need for a database that has all the necessary videos in it. The researcher has used the PSL videos from the FESF project which they recorded to make learning of SL easy for deaf and unimpaired people. These videos have taken after proper permission from the relevant authorities.

Data Development

Data development is one of the major steps in the process of PSL translation tool development. In the current scenario, researcher has all the videos in raw form, which means that videos were not labeled properly. These videos' titles need to be categorized, normalized, tagged, and lemmatized. Researcher has a limited number of videos for the development of tool, and the aim is to provide as much translations that are possible. The steps which performed on data are given below:

Identify Bigrams and Trigrams

There are few words that provide meaning when connected with another word in a sentence. For example, "pass" and "port" become different words when tokenizing a sentence, but these two words should come together to convey proper and accurate meanings. In a sentence, there are a lot of examples of such kind of words. These kinds of words can be understood by the humans, but machine cannot translate or understand these words, until proper instructions are provided by the developer of the tool/software. To develop the data, identification of bigrams and trigrams was performed, using the NLTK library mentioned in the previous section. Bi- and trigrams were identified and separated while normalizing the data.

Parts of Speech Tagging

Parts of speech (POS) tagging is another important step of data development. There are different words with the same spellings but with different senses and different POS family. For example,

1. "She saw a bear."
2. "Your efforts will bear fruit."

In first example, bear is a "noun," and in the second example, bear is a "verb." To avoid such kind of ambiguity, there was a need to tag the words according to POS. In the process of preparing the data, POS tagging was done by the researcher.

Lemmatizing

The process of converting the words of a sentence to its dictionary form is called "Lemmatization." For example, given the words entertainment, entertaining, and entertained, the lemma for each and all would be entertain. After lemmatization process, researchers got the root word. Now, "PyDictionary" is used to get the synonyms of the words in the database to increase the number of words.

Video Extension Conversion

A custom script was written to convert the format of the videos, saved in the database. By default, videos were having the ".wmv" extension, whereas the requirement of PSL translation tool was ".mp4". So, the script converted the ".wmv" extension into ".mp4."

Database Creation

After applying all the steps mentioned earlier, a "MongoDB" database was created, and all the processed videos were stored in it. There is a concept of "collection" and "document" in the MongoDB, document exists within the collection. Videos are stored in the "collection" according to POS tagging, and records are saved in "document" which consists of the fields given below:

a. Word
b. Lemma
c. POS Tagging
d. Video Link
e. Synonyms

f. Video Extension
g. Created at

In the MongoDB, these fields look like the one in JSON format. The real example from existing database is given below:

{
"_id": ObjectId("5a8d6d17b5966f22a4a09bd2"),
"word": "collage,"
"pos_tag": "NN,"
"video_link": "f:\\psl data\\psl\\arts\\collage.mp4,"
"synonym": "photomontage,abstract composition,found art,"
"video_ext": "mp4,"
"created_at": ISODate("2018–02-21T12:59:03.853Z"),
"lemma": "collage"
}

Text to PSL Conversion

The Natural Language Toolkit (NLTK) is used to convert the text to the PSL. The final output of the previous steps is the tokenized form of written text. As mentioned by Khan et al. (2015), like all other SLs of world, PSL also has a specific grammatical structure. At this stage, text is converted to valid PSL syntax form. English sentences are manipulated with the help of NLTK written in Python programming language using the libraries mentioned earlier, that is, identification of bigrams and trigrams, POS tagging, and lemmatizing.

English to PSL Clips Matching

As researchers are using the prerecorded videos of PSL, so after the conversion of English text to PSL, video matching mechanism starts. A custom algorithm matched the PSL prerecorded videos with the sentence written or spoken by the user. This algorithm matches the video for each word from the already developed database. It starts when the user speaks or writes an English sentence in the text area of the PSL translation tool. The user writes a sentence, the algorithm starts from here, and it tokenizes the sentence. Once the tokens are generated, it removes the "stopword" from the tokens, except the question mark "?" and the exclamation mark "!" because in few sentences, the user writes these marks. After this step, the output is "words" generated by the previous steps. Now, the word matching mechanism starts at this step of the algorithm. If the words are matched, it simply provides a video link and displays it, but if the words does not match, it further proceeds and starts the process of lemmatizing the words. Once the lemma of each word is obtained, it checks the database, if the lemmas are matched with the lemmas in the database, it provides the video link. If it

does not match the lemma from the database, the algorithm further proceeds and finds the synonyms of previously generated words and tries to match with the database. On matching of the words with synonyms in the database, it provides the video link; if not, it finds the synonyms for each lemma and provides the video link. After all these steps, if the words or lemmas still do not match with the database, it shows the message that the word does not exist. A detailed diagram of this algorithm is mentioned Figure 2.

Clips Binding Mechanism

Once the clips are selected by the abovementioned algorithm, now there was a need to bind all the clips together to show it as one video clip. As there are large number of prerecorded video clips in the database and each word has one video clip, this stage selects all the relevant clips from database and process them with the help of "MoviePy." "MoviePy" is a Python module for video editing, which can be used for basic operations (such as cuts, concatenations, title insertions), video compositing (a.k.a. nonlinear editing), video processing, or to create advanced effects. It can read and write the most common video formats, including "GIF." It works at the backend of application. In the PSL translation tool, there is a single video for a token (word). MoviePy, first of all, resizes each video and then concatenates those processed clips. In this tool, the resolution of each video is "480 × 320." After equalizing all the videos, it binds and processes it as one video clip.

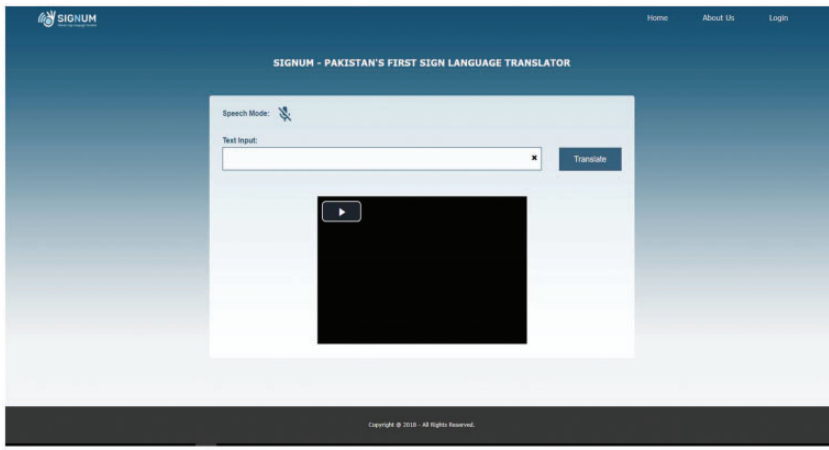


Figure 3. The main page of the PSL translation tool.

Video Output

After successfully completing all the abovementioned steps, a final video is formed. Now, there is a need to display the video clip in the browser to convey the message to the deaf people/students. To display the video in the

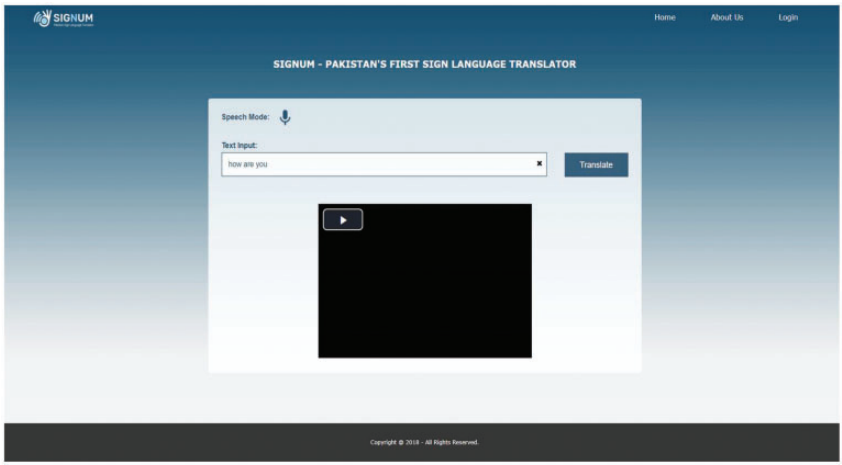


Figure 4. Speech to PSL translation video output.

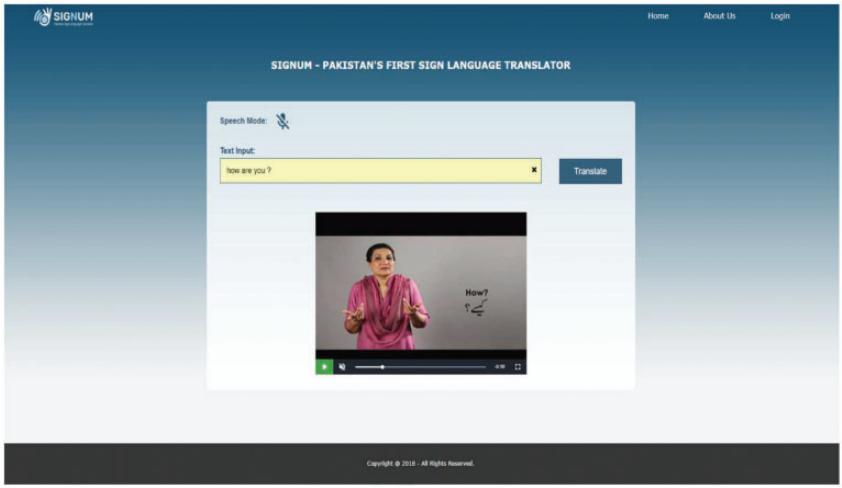


Figure 5. Text to PSL translation video output.

browser, researcher is using the "video.js" for this purpose. "Video.js" provides a common controls' skin built in HTML/CSS, fixes cross-browser inconsistencies, adds additional features such as full-screen and subtitles, and manages the fallback to Flash or other playback technologies. This final video covers all the words written in a sentence to form a meaningful PSL sentence. This final video also has English and Urdu subtitles which helps the deaf or hard of hearing people to learn vocabulary as well. The screen shots of the PSL translation tools are mentioned in Figures 3 to 5.

Conclusion and Future Work

The researchers have provided a literature review of work done by the several researchers in the world and in Pakistan as well to bridge the communication gap between hard of hearing and unimpaired people. The literature review also showed that such kind of translation tools with bilingual subtitles (English and Urdu) can be used in educational settings of Pakistan to teach the deaf students and to enhance their vocabulary with the help of subtitles. In all these works, technology has the main role to design and develop a tool to ease the communication between deaf and unimpaired. People have worked on different ideas, all across the globe, to bridge this communication gap. In Pakistan, researchers have started working on this area also, but there is no any significant work yet. This research has highlighted this gap and developed a prototype to translate the English text and speech to PSL to bridge the communication gap between deaf and unimpaired people.

In future, the researchers will also develop a fully functional tool to translate English text and speech to PSL and make the framework supportable for multiple platforms, that is, web and mobile based. To determine the viability of this tool in educational settings, surveys and interviews will be conducted to collect the data from the teachers who are teaching deaf students in separate classes or in the mixed classroom where deaf and unimpaired students study together and also from the domain experts. A survey will also be conducted to determine whether bilingual subtitles (English and Urdu) can be helpful to teach the vocabulary to deaf students or not.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

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